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VOL. II.

MARCH, 1892.

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*** START OF THE PROJECT GUTENBERG EBOOK THE
AMERICAN BEE-KEEPER, VOL. II, NUMBER 3, MARCH, 1892 ***

THE AMERICAN BEE-KEEPER



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March, 1892.

NO. 3.

Hints to Beginners in Bee Culture.

BY H. M. DEWITT.

This is the month that we should begin to feed and build up our bees, especially our weak colonies, and to get them ready for the honey harvest. Commence by giving them one half pint thin sugar syrup each day; do not feed them in the daytime, feed them at night and they will have all the feed taken down before the next morning. This will start them to rearing brood rapidly and by the time the honey harvest arrives they will be strong and overflowing with bees ready for it. Make but a limited number of swarms and make them strong and early. Late natural swarms should be returned to the parent hive, about twenty-four hours after hiving them. The colonies that work freely on red clover should be used as breeders in preference to others as the tongues of these bees are evidently longer.

The old queen always goes with the first swarm unless she is unable to fly. When making artificial swarms raise your queens and drones from the best colonies. A queenless colony will raise queens at once if it has larvae less than three days old and these queens will hatch within 10 to 12 days. If you give your bees a good supply of empty combs before the beginning of the honey crop and keep them at work they will rarely swarm. But if they once find themselves crowded and get the swarming fever, nothing will keep them from swarming. The honey harvest lasts but a few weeks, so you must be ready for it. "Make hay while the sun shines." When hiving a swarm give them a hive full of worker comb, or comb foundation if possible, or else give them narrow stripes for guides, but do not give them a hive partly filled with comb, as they would be sure to build a great deal of drone comb in the remaining space.

BEE DIARRHEA, FOUL BROOD, ETC.

Bee diarrhea in the latter part of winter and early spring is a malady that effects some apiaries. The bees discharge their excrements over the hives and combs, producing a dark appearance and offensive odor. The cause is either fermented honey, improper food, long confinement, or too warm and poorly ventilated quarters. Give them good capped honey and a cleansing flight. If too cold for this out-of-doors take them into a warm room, make a box, with the front and top made of wire cloth, or mosquito netting, adjust it to the entrance, so that the bees must enter it on leaving the hive. This will usually prove an effectual remedy.

FOUL BROOD.

Foul brood is the rotting of brood in a hive; the caps of the sealed brood appear indented and shriveled and the larvae and young bees in unsealed cells become putrid, emitting a disgusting stench or smell. When the disease has a firm hold, even though it may be possible to cure it, I would advise the total destruction by fire of the bees, combs, frames and hives, with everything which might harbor the disease. In its primary stages it can be cured in this way: With an atomizer spray the hives, bees, brood, honey and combs with a solution of salicylic acid, borax and rain water, repeated on the sixth day. Remove the diseased brood from the hive and give them capped honey, if not too far advanced this may give relief.

There is another plan, which is as follows: Take a clean new hive with new, clean frames, fill it with comb foundation, take and run all the bees out of the diseased hive into the clean one, do this in the evening and as soon as the bees are all in close the entrance with wire cloth, keep them confined for forty-eight hours until they have consumed all the honey in their sacks in building comb. At the end of forty-eight hours open the entrance and let them fly if they wish, feed them a little sugar syrup every night for about a week, and if the honey season is over, or, if this is done during a dearth of honey you should feed them regularly so as not to let them starve. I had the disease in my apiary the past season and this is the plan I used to cure it. My bees are as healthy now as as if they had never had it.

Sunny Side, Md.

[The instructions which friend Dewitt gives in the first part of the foregoing article will apply this month only to the more southern localities. Here in the North the hives in many places are still covered with snow and the bees should not be disturbed until spring has unmistakably arrived.—ED.]



New Inventions.

BY JOHN F. GATES.

The question has been asked “Are we drifting from our moorings.” I used to think that we should not, but if all bee-keepers anchored to one idea there would be no improvements. While it is safe to our own pockets to be conservative, yet no class has done more to advance the interests of the bee-keepers than those who experiment, and seem not to be satisfied with their present condition. Had the inventors of the Monitor been contented with wooden war ships our great American Republic would have been divided. Had we all been content with stage coaches where would our railroads have been? Had Edison preferred to sit at his telegraph instrument we should now be without his master ideas. This onward impelling force in Americans has sought out so many good things in the last fifty years that I have not space to tell them. Some rejoice in real improvements. Well, we can’t grind out out a grist of real improvements to order. We have many discouragements and losses before we succeed in turning out one. Many of these inventions must be tested by bee-keepers before a true verdict can be given, and we should all be willing to lend a hand to be one of the great jury in the discussion of these cases as they are brought before us by our leaders; the inventors. Yet while the tester goes hand and hand with the inventor, each watching the others movements, each helping the other to discover and rectify mistakes. It is too true that many good inventions have been swamped and for years laid dormant when they might have been in use, simply for the lack of wisdom to guide us to small experiments first. Yes, there seems to be too much rush, new things can’t be tested in a hurry. To change an average apiary all at once to some new mode of management, or new style of hive, even if the hives were given to us, would be unwise. But add the cost of hives and fixtures which the change involves with the loss which one is sure to meet with for a time under any new arrangement, and can we wonder that there is so little confidence placed in inventions or the

inventors. Still had we gone more slowly, tested more carefully, and on a smaller scale and given ourselves more time to sum up the evidence, no doubt many times our verdict would bless instead of curse the inventor. No doubt there are inventors who abuse one's confidence, but they too well have but little chance to deceive us if we go slow. We can change too much, and again too little. I am aware that I have missed some good opportunities by being a little too set in my ways, and I have had too little charity for improvements; medium ground is safe ground on which to stand. We should watch the signs of the times and not jump conclusions, nor bite at all that takes our fancy, nor kick at all that we despise, we ought always to review, draw conclusions and watch very closely what the mass of bee-keepers seem to favor, or decide upon. If we are good readers of indications we need never get left, and often can go across lots, thus reaching the head of the procession, but be sure we know the way across else better we had gone around.

Ovid, Erie County, Pa.

What I Have Observed, Etc.

BY T. K. MASSIE.

(Concluded.)

In the last article I promised to give some evidence confirming my conclusions, but before doing so I want to say a word in regard to “large vs. small hives.” In the discussion of the subject in the different bee journals from time to time, I neglected to note the size of the frame used by the advocates of a large hive, but I imagine they use a deeper frame than the “L,” and if so they are evidently right in advocating a large hive, for such a hive would be better proportioned, and would conform nearer to the natural requirements of the bees than a small hive with shallow frames.

In *Gleanings* for July 15th, page 553, friend C. J. H. Gravenhurst, in speaking of “handling hives instead of frames,” hits on the same ideas given in these articles in regard to the winter problem. He tells us that the bees winter better in the straw skeps than they do in the movable frame hives, as made and used at present. This is because the bees in the skeps have their hives propolished overhead which prevents upward ventilation and keeps the bees dry. He also says he gets more honey with less labor and cost; then he shows how he sought to combine the skeps with the movable frame hive, advocating about the same advantages that I have given in these articles.

But the most clinching arguments in favor of doing away with the useless Hill device &c., is found in Ernest R. Root’s review of G. R. Pierce’s book, “The Winter Problem in Bee-Keeping,” which appeared in *Gleanings* for December 16, ’91, page 952. Mr. Pierce says the pollen theory is not the cause of diarrhea; that diarrhea in bees is caused by cold and lack of stores, and is only intestinal catarrh.

Chaff cushions, or other porous material over a sealed cover are all right and serve a good purpose.

Mr. Pierce is a thorough advocate of protection and packing around the bees; but the cover must be sealed down that no heat can escape into the packing above. In the first of this series of articles I took this same position. I said “Therefore I have drawn the conclusion that a thin walled hive, protected by a movable winter case, and packed on all sides with a cushion made of felt and filled with some non-conducting material—one that will prevent all radiation of heat will be best”—and, in substance, that we could remove cases and packing on warm days and have our hives purified by the sun and air and protect our bees by wrapping them up with the warm cushions and prevent the radiation of heat at night. In the second article (See AMERICAN BEE-KEEPER, page 164) I said the “pollen” theory, and upward ventilation, cut no figure in the winter problem in my locality. Now if we place a thin board down solid on the top bars of our brood frames early enough in the fall for the bees to glue up the crevices and thus prevent all air currents from passing up through our hives, we again get even with our box hive brethren, and when we prevent all radiation of heat by placing a cushion on top of this board, the same as we do the sides. And further, in spring and early summer, when we give our bees just the ventilation required by raising this board, we are another long step ahead of them.

In the winter of 1891 I had The W. T. Falconer Manf’g.’ Co. make for me closed end frames with winter passages through the top bars, and boards to be sealed down for the purpose of carrying out my plan as given above.

“Ernest” tells us of his experiments in using thin boards and pieces of glass imbedded with white lead paste, as it was too cold for the bees to seal them down with propolis. Under the glass he placed a thermometer, which, when the weather outside in the wind was ten degrees above zero, registered 45 to 50, and “*the hive was perfectly dry inside.*” These are valuable experiments in the right line.

My ideas, as it is plain to be seen, were given to the public before friend Pierce’s book made its appearance, and before “Ernest” tried any experiments on this plan.

On page 592 of *Gleanings*, L. Stachelhausen tells us that having found out the advantages of closed end frames, he will use no other. The closed end frames have only to be given a fair trial to prove their superiority over all hanging frames. All the “rattle traps and nuisances” I have mentioned in these articles will soon give way to something better and more simple.

Friend Lowry Johnson also is of my way of thinking, as his article in the *AMERICAN BEE-KEEPER* for December, '91, page 182, will show. Also Brother Quigley of the *Missouri Bee-Keeper*, is advising his readers that a board sealed down on top of frames is better for wintering than cushions next to the bees. See his answer to a correspondent to his paper, page 144.

These articles end here, but I would like the opinion of the reader on the points taken in them.

Concord Church, W. Va.

A Talk on Bee Hives—Fixed Races—Honey Crops, Etc.

BY S. L. WATKINS.

The production of honey is the principal object that a beginner has in view when he contemplates starting in the apiarian business. Of course, he generally buys a few box hives and does the transferring himself; which I think is a good idea, as he gains considerable knowledge of the inside of a bee-hive and of handling bees. Before he invests in bees, he generally buys and reads a couple of bee books and obtains a few catalogues of leading apiarian manufactures, to see what style of hive is best. If he is gifted with an average insight into the mystery of common things, he will quickly choose the hive and system that are most universally used, and will stick to that system, and nine chances in ten he will make a success of his bee business. If, however, he is gifted with a volatile nature, he will not be satisfied until he has eight or ten different style hives in his apiary at one time, and will spend all the money he makes in trying new apiarian fixtures, until he finally gives the bee business up in disgust. I do not mean to say that all new bee hives are useless—far from it; but generally speaking, there is a flood of new style bee hives on the market which are miraculously complicated and contain numerous paraphernalia in the way of wedges, glass doors, clasps, useless bee spaces and other ornaments not worthy of mention.

A hive that is too complicated will never come into general use. Competition in the honey business requires that we use the cheapest appliances, combined with the greatest excellences. Here in this state (California) where honey is so cheap, it would be folly to spend \$8.00 for a bee-hive, because a \$1.50 hive will answer every purpose equally as well, the hive does not make any difference in the amount of honey gathered,

bees will store as much honey in a box hive as in any frame hive ever devised; the queen and race of bees make the difference in the amount of surplus gathered. The simplicity hive, with its various modifications, is the hive that gives the best satisfaction among advanced apiarists, and when used with the Hoffman frames it is hard to beat. To persons who contemplate starting out-apiaries, the Hoffman frame offers very superior advantages. My opinion is, that bee-keepers who keep out-apiaries, and who move bees considerably, will in time settle down to the fixed frame. I shall no doubt experiment considerably with fixed frames the next few seasons.

In Ventura county this state, the bee-keepers have adopted the Langstroth as the standard frame, and there is something like 1,600 hives in that county, which produce annually about \$60,000 worth of honey. The one pound section is rapidly gaining favor with the progressive apiarists of this state, and are fast superseding the old Harbinson two pound section.

Our honey crop here last season was about one fourth of a crop. In the upper Sierras, at an elevation of 3,000 feet and upwards, there was a good crop. Some honey plants yielded well. In all extensively irrigated districts bees did pretty well because of the abundance of alfalfa grown. Reports from Antelope Valley state that in that section the honey crop was far better than usual, 200 to 400 lbs. to the colony for entire apiaries being the yield.

The Italian race of honey bees I have tried pretty extensively, and found them to be very good, but I like the Carinolans better. I think they are a fixed race; the Italians are not.

In an apiary composed of Italian and hybrids if a Carinolan queen be introduced and the Carinolans then be left to reproduce themselves naturally they will hold their own for hundreds of generations before their markings will begin to be eliminated. Place a colony of Carinolans in an isolated location, and allow natural breeding, and in ten years they will not deteriorate a single bit; but take a colony of Italians, and allow natural breeding and in a year or two we have nothing but common black, or very poor hybrid bees; thus proving conclusively that the Italians are not a fixed race.

Grizzly Flats, Cal.

A Few Words to Beginners.

BY T. K. MASSIE.

AS THE AMERICAN BEE-KEEPER is published in the interest of beginners, allow me to say to them that the first thing to be learned is the fact that no set rules can be given to suit everyone under all the varying conditions of climate, location, etc., and that everyone must use intelligence and make rules to suit his own individual case—must make a vigorous use of his own “think shop.” He must thoroughly acquaint himself with the flora of his location; he must know when the honey flow is likely to commence and end, and must manipulate his bees to suit that time. Friend Doolittle’s advice to manipulate our bees at the proper time cannot be too strongly impressed upon the minds of beginners. In my locality we have two honey flows each year. The early flow last season ended on the 24th day of July, the fall flow commenced on the 16th of September. There was a dearth from July 24th to September 16, of 53 days. A hive full of bees during these 53 days are not only of no use to me but a positive disadvantage. They are only consumers and not honey gatherers because there is no honey to be gathered. Reason would dictate to me that I must use every means in my power to build up my colonies as strong as possible from early in the spring to within 35 days of the end of the honey flow, or about the 19th of June, that about this time, certainly not later than June 25th, I must restrict my queen to as few frames as possible, so as to have but few bees during the 53 days of dearth. (I base my calculation upon the fact that 21 days are required for the egg to hatch and then the bee must be 14 days old before it becomes a forager.) Now I wish to say emphatically that every one whose location is similar to mine must practice restriction or his honey crop will be a failure every time. Again by the 16th of September. I want my hives as full of bees as possible ready for the fall flow. Then going back 35 days from September 16th takes me to August 12th, the day on which the eggs must be laid for the bees to have hatched out, ready for the beginning of the

fall flow. But as the queens could not under any circumstances fill the hives full of eggs on the 12th day of August, (or any other one day for that matter,) reason would again tell me that I must remove the restrictions from my queens two or three weeks before, or, say July 24th to August 1st, and then stimulate brood rearing. If the beginner fails to follow this plan he will certainly get no surplus fall crop. We must also make sure that each colony contains a good prolific queen.

Brother Demaree's "Practical hints in Bee Culture" on page 179, *AMERICAN BEE-KEEPER*, is timely and should be read and studied until perfectly familiar with every "hint" he has given. When he speaks of the bees "crowding the queen" being simply an effort on the part of nature to assist the bees by curtailing brood rearing during the honey flow he is certainly correct, yet, by restricting our queens as I have suggested above we accomplish the same end, provided the restriction is performed at the right time. When he speaks of giving his colonies, after swarming, a queen cell instead of a laying queen, as some "innocent bee men" had written him, he gives a "hint" that should be well and long remembered. By this plan he also assists the bees in curtailing brood rearing, when such brood would hatch out bees that would be consumers instead of honey gatherers.

His article on "Concentration of Forces" in *Bee-Keepers Guide* for December, 1891, page 356, is worth the price of that paper for several years. "Concentration of forces" is his remedy against poor honey seasons. His plan of "concentrating forces" can be easily carried out in connection with the suggestions I have given. He says "not a swarm should be allowed to issue." He accomplishes this by one single manipulation, which is simple and easy, and performed "just at the commencement of the honey season, and before any swarms issue." All the colonies strong enough to cast swarms are treated on the following:

PLAN.

"All the combs containing brood are removed from the brood chamber, except one that contains but a small amount of unsealed brood and eggs. This is left in the brood chamber with the queen on it. If she is not found on

it she must be hunted up and put upon this comb. The brood chamber is now filled out with empty combs and a queen excluder is placed on its top. The combs containing brood are adjusted in a super or hive body, and if they do not fill it, it is filled out with empty combs. It now goes on top of the brood chamber with the queen excluder between. We now have all the brood above the excluder, except what is in the comb with the queen on it below the excluder. You now have nothing to do but to “turn up” to suit the season. Treating all colonies in this way the season will have to be more than usually extended if there is a single swarm. Colonies treated in this way are the strongest colonies I ever handled, and I never seen a season so barren of nectar that they fail to fill the combs above the excluder by the time all the brood they contain are hatched out, and if the season is a good one they will surprise the natives and make you uneasy about the safety of your honey floor, like mine did me the past season, though the season was but an average one.”

Next month I will give my experiments with two colonies the past season managed on a plan similar to to Brother Demaree’s, as given above. Remember, beginner, that if you don’t manage your bees intelligently you had better never go into the business for you will have no “luck” and the business will be a failure with you; and you might as well try to raise corn without seed as to try to raise honey without a scientific work on bee culture and one or more of the many excellent bee journals.

Concord Church, W. Va.

(Concluded next month.)



OUR CORRESPONDENTS

ED. AM. BEE-KEEPER, Dear Sir: I inclose herewith 50 cents for renewal to the BEE-KEEPER, and in a month or so you will receive a little order for supplies from me, but I wish to find out first what I need. There is a lot of reading in the AMERICAN BEE-KEEPER for beginners, but there are so many different ideas given that the beginner must have a big head indeed to remember them. I will take the liberty to tell you how I began. It was a year ago about Christ Christmas when I resolved to keep bees. I wrote to the *New York Volkszeitung*, a German newspaper, for the address of a good bee paper, and I received your address. The same day I sent 50 cents and received in time the *American Bee-Keeper*, in which I found the address of Mr. Knickerbocker of Pine Plains, N. Y., to whom I wrote for a catalogue and price list for bees and queens. The price of a good colony of Italians was \$8.00, and I sent him \$16.00 and got two colonies of bees about the 20th of May, which was after all fruit trees had blossomed. The bees commenced to work, and the 15th of July I took out of each hive the surplus case of 28 sections well filled with white honey. I replaced them with new surplus cases, and about the 12th of October both the hives were as full of honey as they could be again. I left the hives in this condition for winter so the bees can consume as much as they please, and if there should be any

honey left this spring it is just as good for me as if I took it out in the fall. I left them on the summer stands packed up good and warm and will leave them alone until spring. I do not think much of looking into the hives every little while. I may be wrong, but I think this would make the bees swarm earlier next season.

I cannot find any one around here who knows anything about bee-keeping. I sold 40 sections of honey and 16 I kept for myself. There is at least 35 pounds in the hive yet, and I think I did very well for the first year. With regards I remain,

Yours truly,

F. TLEGMAN,

Seymour, Conn., Feb. 10, 1892.

[FRIEND TLEGMAN: We are glad to hear from you and your success the first season with bee-keeping. You were unusually fortunate with your bees and must not be discouraged if the coming season or the one following, you find your bees doing nothing. Evidently you are in a good location for keeping bees, and would advise you to go into it more extensively. We shall be pleased to hear from you again later in the season.—ED.]

ED. AM. BEE-KEEPER, Dear Sir: As my subscription has expired, I send you herewith a postal note for 50 cents, as I cannot do without the BEE-KEEPER. I think it is the best bee paper for new beginners or old ones. I put seven colonies in my cellar and left the balance on the summer stands, so far they have all done well; some that were a little short I fed up. Last season was a poor season for bees here, so those that came off late did not gather enough to get through the winter.

My way of feeding bees is to get an empty comb and put it on a thin board, then put it on the top of the hive and put sugar syrup in it. The bees will fill themselves and carry it and put it in the empty comb below. Yours truly,

JOHN W. HARRIS,

Colfax, Ind., Feb. 8, 1892.

ED. AM. BEE-KEEPER, Dear Sir: I have neglected sending in my subscription but will do so now. You will find inclosed \$1.00 to pay arrearages and also for another year's subscription. We like the BEE-KEEPER very much and think every bee-keeper should take it.

The winter of '90 and '91 was a hard one in Maine for us bee men. I lost all but three swarms. Did not get a drop of surplus honey; something that has not happened before in the forty years we have had bees. Up to date what few I have appear to be getting through the winter all right. The weather has been quite mild with but little snow.

Yours, &c.,

EZRA WITHEE,

Pittsfield, Me., Feb., 8, 1892.

ED. AM. BEE-KEEPER, Dear Sir: I think the AMERICAN BEE-KEEPER is the best paper for the novice there is published. It does not publish things unconcerning bee business, like others, as for instance, in regard to tobacco, gardening, &c.

I will tell you how I pack my bees for wintering. I had 12 strong colonies in ten frame simplicity hives about Nov. 20th, I did not molest them in the brood chamber at all. I just took off the top of the hive and set on top of the frames a super without any sections in it and filled it with buckwheat chaff and packed it tight. To be sure the enameled cloth was on top of the frames under the chaff. There was enough ventilation between the super and the side of the top of hive or upper story, as I do not think the caps on the hives were any too light to prevent the foul air from escaping. I will tell you in the spring what success I will have had in wintering.

Yours truly,

OTIS CALLAHAN.

Wellsboro Pa., Jan. 25, 1892.

ED. AM. BEE-KEEPER, Dear Sir: Bees are wintering well in Michigan; at least mine are wintering well on their summer stands. Had a good flight February 12th. In looking through them I find they have plenty of stores. My crop of honey last year from seven colonies was very good. I increased to twenty colonies and got 450 lbs. of comb honey besides, in one pound sections. This is my third year in the bee business and I like it very well and intend to keep at it.

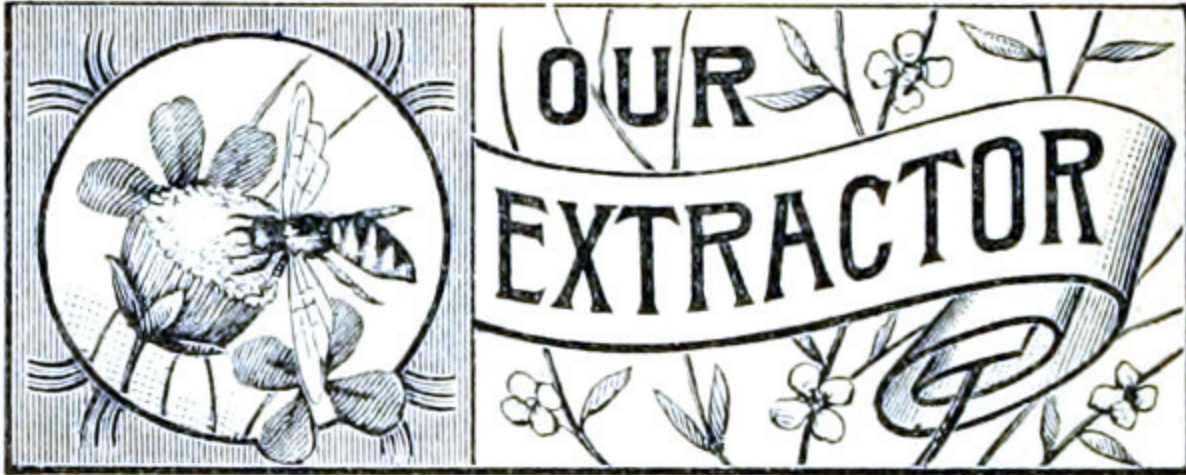
G. W. FRANKS

Belzes, Mich.: Feb. 20th, '92.

ED. AM. BEE-KEEPER, Dear Sir: We have watched every day for my BEE-KEEPER because I am anxious to see it. Papa got twenty hives and 1,000 sections from your factory last year. He thinks the outside winter case is very nice. He can put a warm brick in under the cushions and feed the bees almost any time in the winter. He has the outside cases painted red on the sides and ends, and the top is white. He faces them to the east so the morning sun will warm them up and get the bees up early in the morning. Yours truly,

OLLIE JONES.

Cortland, O., Feb. 14, 1892.



OUR EXTRACTOR

WINTERING ON SUMMER STANDS.

The question of wintering is always of interest and for that reason is always seasonable. In treating the question I do not expect to give any new points to experts, but many of the readers of this magazine are beginners, and are looking to this valuable journal for information on this as well as on all other apicultural matters. Many successful apiarists of large experience claim that all wintering should be done in cellars, or other special depositories; with these I have no quarrel, but never having wintered except on summer stands, I am unable to speak thereof from personal experience. One objection to this plan of wintering I can well imagine will force itself into the minds of a large majority, viz.: the expense required to fit up as it should be fitted, with regard to ventilation and temperature, such a special depository as is necessary in order to guarantee success and the objection of itself will probably prevent that majority from incurring the required expense.

For the benefit of the same majority I will give in detail the plan of wintering on summer stands, adopted by myself with perfect success, and

which I have made use of for eighteen years or more, and that too on Langstroth frames, with single walled seventh-eight inch pine hives.

As theories in regard to matters connected with apiculture are of little consequence when compared with actual facts. I will not attempt to theorize now, but will detail the facts for the use of any who may desire to know them.

When the honey season ends, which with myself is about the 10th of September, I examine the condition of every colony, crowd each colony on to seven frames I intend they shall all be strong enough to cover fully that number and see that each frame is at least filled with sealed stores in its upper half the whole length. Later on when I get ready to back for winter I extract if necessary from those combs that are more than two-thirds filled with stores and combs throughout the brood chamber are equalized and placed in a position where the colony can at times get at them if desired. When the temperature falls so low that the colony begins to cluster closely I force the cluster to one side or other of the brood chamber, which can easily be done, by moving the frames on which the cluster is formed. Prior to this, however, I have stimulated the queen by feeding regularly each day a small amount of sugar syrup, and thus kept the colony rearing brood as long as possible.

After the cluster is forced to the side of the hive I place a "Hill's Device," or some substitute therefor, over the frames, and cover the bees with a light porous blanket. Burlap or cotton duck is as good as anything for this purpose. The "Hill's Device" under the blanket forms a means of communication for the bees with every frame in the hive, and that too without danger of becoming chilled. As the hive in use is wide enough for ten frames I use one and a half inch division board in each side of the hive, which allows the seven frames to be spread apart a little more than desirable for summer use. After covering the frames in closely so that not a bee can show his head outside, I put on an upper story and fill it one-third full of forest leaves pressed lightly down, and use a cover with one and a half inch hole bored in each end for ventilation.

I give a large entrance, using a bridge about four inches wide for the bees to crawl under, which prevents the easy access of sudden draughts into the

hive. The only other protection than that prescribed above, found in my apiary is a close osage hedge, six feet high on the north and west sides. With the above means of protection my bees have withstood the rigors of our eastern winters for years, with a temperature varying from 20 degrees above to 20 degrees below zero, and some seasons without a purifying flight from middle of November to middle of the following February. Many mornings with the thermometer below zero in January I have found a warm current of air being forced out from the entrance, so strong as to be perceptibly felt upon the back of the hand. I know not and care not whether others may agree with me or not, I state the facts as I find them, and have no hesitation whatever in advising every beginner to follow the methods outlined above.—*J. E. P. in B. K. M., (Mass.)*

SURPLUS HONEY.

Under the modern system of bee culture we obtain two kinds of honey, known as comb and extracted. Comb honey is brought before the consumer just as it was stored by the bees, while extracted honey is the pure honey emptied from the comb by means of the honey extractor. If not adulterated by middlemen both are equally healthful and nutritious.

Honey is not a luxury, but a necessary addition to our food, it being the pure sweet as secreted by the flowers from which it is gathered by the bees, being healthful and much safer than the poisonous confections sold under the name of “candy.”

Comb honey is preferred by many on account of its fine appearance. It must be placed in the market in good shape, indicating that it is intended for food and not simply as a luxury, a sweet morsel to be tasted by the children. Some of our writers on bee culture (but I am happy to say only a few) went wild some time ago advocating one-half pound frames. They argue that it can be sold at a higher price, but all bee-keepers know that forcing bees into such small combs greatly reduces the crop, and if such a course could be pursued by our bee men it would at once convey the idea that honey was only a substitute for candy or chewing gum. My advice to the bee-loving readers of this magazine is to let them severely alone in their craze. A

season or two will abolish such small things. The man whose soul is so small and his ideas so contracted, and his business principles so mean as to place a half pound honey package upon the market for the purpose of extorting a half penny or so from his fellow man deserves to be classed with—you may draw your own comparison.

Anything smaller than a one pound section or frame is a loss to the producer as well as the consumer. A one pound comb makes a nice package, and such frames can be nicely crated and safely shipped.

Comb honey should be removed from the hive as soon as all the cells are sealed over. If left to remain it becomes darker by the bees passing over it. When taken from the hive it should be placed in a dark room until sold or shipped to market. Some writers advise smoking it with sulphur to kill the wax worm. I never found this necessary, as I never have found worms on my comb honey. Honey should be nicely graded, and the finest shipped or sold in separate lots. My frames hold two pound each, of these I place six in a crate having glass at each end showing quality of honey. These crates suit the retail as well as the wholesale dealers and consumers generally buy a whole crate—12 to 13 pounds. Many of my customers in adjoining towns buy from three to six crates. Unfinished or partly sealed combs can be emptied with the extractor and put away for next season.

Extracted honey should be placed in nice, clean, attractive packages. For home trade self-sealing jars do very well. For shipping, kegs are found to be the best.

Just here I would give a word of warning. Do not extract before your honey is sealed, if you do it is not ripe and too thin and will sour, thus spoiling your reputation as a honey producer. Don't be too eager to obtain a large quantity, let the quality be good and you will have no trouble to find plenty of buyers. It is only the poor, unripe article that cannot be sold, and which gives some of our writers in the bee papers so much trouble to find a way to dispose of their honey.

(Penna.)

H. H. FLACK.

EXPOSE YOUR HIVES TO THE SUN.

In a village in Germany, where the number of bees kept was regulated by law, a bad season had proved that the place was overstocked, from the great weakness of all the colonies in the neighborhood. There was but one exception, that of an old man who was generally set down as being no wiser than his neighbors. The honey harvest came round, and when he had stored away double the quantity that any of the rest had saved, he called his friends and neighbors together, took them into his garden, and said: "If you had been more charitable in your opinions, I would have told you my secret before," and he pointed to the facing of his hives—one degree more to the east than was generally adopted. The sun came upon his hives an hour or two sooner by the movement, and his bees were up and stirring, and had secured a large share of the morning's honey before his neighbors' bees had roused themselves for the day.

SNOW AND OUTDOOR WINTERING.

The heavy fall of snow which we have just had is very favorable for outdoor wintering, and we would advise our readers not to shovel it away from the hives, but if you have time put a little more on with a snow shovel. We would much rather have ten feet of snow over hives than none at all.

A friend of ours once told us how he had made a great mistake by digging away the snow from some of his hives, but at the same time learned a valuable lesson. His apiary was situated in somewhat of a valley, and one morning after a heavy fall of snow which had been considerably drifted, he looked out and was dismayed to see his entire apiary buried beneath the fleecy flakes. One portion, however, was much worse than the other—that at the north end being buried in some places ten feet deep. The south end was not so bad, and so he determined to clear what he could and leave the rest to perish, as he supposed. After considerable shoveling he got about 25 colonies pretty well cleared off, and by dint of hard work managed to keep them clear till spring. When warm weather came these 25 colonies were flying in and out, while the snow was still lying upon the other portion. Thinking the latter were dead he paid no attention to them till quite late in

the spring, when the tops of some of them began to show above the snow, and what was his surprise and delight to discover every colony in splendid condition, some of them filled with brood, and all ready for a good season's work, while at the same time they had not consumed nearly so much stores as those he kept clear of snow! Upon investigation he discovered that the heat from the bees had melted spaces about the hives varying in size from a square foot to a square yard, and the air from the hives became purified by contact with the snow, while at the same time the temperature was kept so even and was so little affected by the cold breezes of winter that very little stores were consumed.

The method of purifying air is one which is taken advantage of by some of our native animals. We have often amused ourselves by watching the otter who will stay underneath the ice for hours together engaged in fishing, and when finding it necessary to breathe will place his nose against the ice, expel the air from his lungs when it forms a bubble between ice and water, and then inhale it again.—C. B. J. (*Canada.*)



STRAWBERRIES IN OCTOBER.

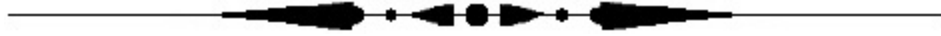
(From Western Garden, Oct. '91.

To-day's mail brings a fine specimen plant of the New Wonder Strawberry from J. B. Alexander, of Hartford City, Ind. It is a strong plant, and the peculiarity about it is that it has three ripe and fourteen green berries on it, besides quite a number of blossoms. Our readers should try a few plants of this wonder. See Golden Rule Nursery "ad." elsewhere.



Some women cannot keep bees, any more than some men; but many can, and to their great profit. Often a farmer's wife or daughter welcomes an occupation for the sake of its novelty, something to break up the routine of cooking, washing and sewing; and bee-keeping, even if it brings only a few

pounds of honey for the table, is undertaken and carried through with pleasure and delight.—*Ex.*



Let it be remembered, says Julia Allyn, that the more bees there are on farms the greater will be the product of the farms; for the bees distribute pollen and fertilize flowers more thoroughly than they can be fertilized otherwise.

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EDITORIAL.

Some of our readers being beginners, are often perplexed to know just what methods of the many different ones advocated by our contributors will be the best for their individual needs. Now, there are many methods of manipulating bees and hives, any one of which followed out will bring success. Take "Wintering" for instance. Some of the most successful bee-keepers winter out-doors. Others in-doors. Some with chaff hives; hives with air spaces or with outside winter cases. Some winter in cellars, and others in special depositories or in bee houses. Each method has its strong supporters.

About the only thing to be considered in adopting either method is the climatic location. For instance, bees in the Southern states will not winter well in cellars, nor are chaff hives necessary, while in the Northern states, outside cases packed, dead air spaces, chaff hives or in-door wintering is a necessity.

A great many letters of complaint have been received from persons to whom we have been sending the BEE-KEEPER, because we asked them to pay for it. Many of them say they never subscribed and do not think they should be compelled to pay. We do not send this magazine to anyone unless ordered to do so, excepting to the former subscribers of the *Advance Bee-Hive* and *Bee World*, whose subscription lists we have purchased, and we have continued after their original subscriptions have expired, excepting when ordered to stop doing so by the subscribers themselves. We have frequently mentioned the necessity of ordering us to stop if the magazine was not wanted, and have sent postal card notices to those whose subscriptions have expired six months back or more. Now, we do not wish any one to take this magazine against their wishes, nor pay for it either, but we do wish you would notify us on a postal card or otherwise, if you want it stopped when your subscription expires.

The Paddock Pure Food Bill now before the United States Senate is one in which every bee-keeper should be interested. The bill provides for the prevention, by government inspection, of the mis-branding or mis-labelling of all articles of food and drugs. In other words, if a can containing honey is marked "Pure Honey," it will necessarily be exactly what the name implies, and not an adulteration.

We have been making an extensive inquiry as to the styles of hives in most general use throughout the country, especially in the Eastern and Middle states, and we wish our friends would send in their views in regard to the advantages of the Simplicity or Langstroth style of hives over the old style box hives. Send it to us either as a regular contribution or correspondence for publication.

We notice in *Gleanings*, *The Review*, *Progressive Bee-Keeper* and other journals the "ad." of the "Chicago Bee-Keeper's Supply Company," in which they state their office as being 68-70 South Canal Street., Chicago. Parties interested have endeavored to find such a concern at this street number but they have failed to do so. A man by the name of Kline, claiming to be the secretary of the company, offered this magazine a very liberal "ad." some months ago, but as our information regarding the concern was not satisfactory we declined to accept the "ad." All bee-keepers will do well to deal only with old established manufacturers and dealers.

Hereafter we will put the name of the state in which they are written at the end of all articles, so that our readers will know that a method or system advanced by a bee-keeper in Georgia, for instance, will not be of much use to any one in this state, especially if it relates to handling bees.

C. H. Dibbern is not satisfied with his bee escape invented the latter part of last season, and claims now that he has another almost perfected which will beat anything yet. A great man on bee escapes is our friend Dibbern. By the

way, M. E. Hastings has recently invented an escape something on the principle of the Porter, which appears to be about perfect. We will endeavor to give an illustration and description of it next month.

Edward R. Newcomb, formerly of Pleasant Valley, N. Y., has moved to Chicago and has given up his supply business; also the manufacture of the Stanley Automatic Extractor.

Everyone whose subscription has expired, or is about to expire, will do well to take advantage of our seed offer given in another column; a \$1.50 box of seeds, &c., for only 25 cents.

The ever-increasing migration to the Tropics from American ports will probably receive a fresh stimulus from the article on the Highlands of Jamaica, which Lady Blake, the wife of the Governor of Jamaica, contributes to the March number of the *North American Review*.

“THE FLOWERS OF CHILI.”

This week's issue of *Frank Leslie's Illustrated Newspaper* being the colored number, contains a beautiful page of portraits of the handsome women of Chili, a sleighing scene in Chicago, character sketches from the tenth annual dinner of the famous Clover Club of Philadelphia, illustration of the Young Women's Christian Association and Margaret Louisa Home of New York City, and of the “Captain Prat,” the formidable Chilian ironclad. The Children's Department contains a beautiful story entitled “Majorie's Valentine,” and the Graphological Department is full of interest, while the fashion letter and editorial pages, together with the beautiful colored front page, make this number the handsomest that the Arkell Weekly Company has ever published. Price only 10 cents; 12 numbers \$1.00, with flower premiums catalogued at \$1.25 by Messrs. Peter Henderson & Co., \$1.25.

The complete novel in *Lippincott's Magazine* for March, "A Soldier's Secret," is by Captain Charles King, who alone among living Americans has the secret of the military tale. What he does not know about army life in the West is not worth knowing, and what he knows he can impart with unsurpassed and unfailing charm. The post, the bivouac, the battlefield,—whatever goes on at these he makes to live again before us; for he has been a part of it all, and his heart is with the cavalry still. His last story has a very recent theme; the Sioux war of 1890,—and will be found equal to any of his previous work.

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ALSIKE CLOVER SEED is considerably higher price now than quoted on page 27 of our catalogue. We can now supply a limited quantity at 25 cents per pound, \$3.00 per peck or \$11.00 per bu. Postage 9 cents per lb. if by mail. Price subject to change without notice.

We wish to call attention to the fact that we can furnish the *Hastings Feeders* to anyone wanting them. They are first class, and in some ways

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Transcriber's Note:

Obvious printer errors corrected silently.

Inconsistent spelling and hyphenation are as in the original.

*** END OF THE PROJECT GUTENBERG EBOOK THE AMERICAN
BEE-KEEPER, VOL. II, NUMBER 3, MARCH, 1892 ***

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